

2017 Lagos State Model Colleges Screening Test Past Questions & Answers

**Paper 1: Mathematics/Quantitative
Reasoning/Basic Science and Technology**

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2017 Screening Test Into Lagos
State Model Colleges

Solutions to Mathematics

1. Circumference of a circle
 $= 2\pi R$

i.e. $176 = 2\pi R$

$$176 = \frac{2}{1} \times \frac{22}{7} \times \frac{R}{1}$$

$$176 = \frac{44R}{7}$$

$$44R = 176 \times 7 = 1232$$

$$R = \frac{1232}{44} = 28$$

$$R = 28 \text{ cm}$$

2.

$$3^2 - 2^2 = (3+2)(3-2)$$
$$= 5 \times 1$$
$$= 5$$

3.

7 tins of milk cost 147K

1 tin ~ ~ will cost $\frac{147}{7} \text{K}$

$$3 \text{ tins of milk will cost } \frac{147}{7} \times \frac{3}{1}$$
$$= \frac{441}{7} \text{K}$$

4.

Area of a circle $= \pi r^2$
but diameter $= 2r$

$$0.7 = 2r$$

$$\therefore r = \frac{0.7}{2} = 0.35$$

$$\therefore \text{Area of a circle} = \frac{22}{7} \times (0.35)^2$$
$$= \frac{22}{7} \times \frac{0.1225}{1}$$
$$= \frac{2.695}{7}$$
$$= 0.385 \text{ cm}^2$$

5.

Let the number be x

$$\text{Average} = 10 = \frac{17+13+6+x}{4}$$

$$\therefore 10 \neq \frac{36+x}{4}$$

$$36+x = 10 \times 4 = 40$$

$$36+x = 40$$

$$x = 40 - 36 = 4$$

$$\therefore x = 4$$

6.

$$\begin{array}{r|l} 2 & 36 \\ \hline 2 & 18 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$\therefore 36 = 2 \times 2 \times 3 \times 3$$

7.

$$2 \times 8 = 16$$

$$8 \times 8 = 64$$

$$\therefore 2:8 = 16:64$$

8.

$$\frac{1}{3}, \frac{1}{2}, \frac{2}{5}, \frac{1}{4}$$

$$\frac{20, 30, 24, 15}{60}$$

$\Rightarrow$ Re-arrange in ascending order

$$\frac{1}{4}, \frac{1}{3}, \frac{2}{5}, \frac{1}{2}$$

9. A

10.

3,000,000 - Three million
500,000 - Five hundred thousand
96 - Ninety-six

$$\underline{3,500,096}$$

11.

1 meter cost ₹800

$$10\frac{3}{4} \text{ meter cost } 10\frac{3}{4} \times ₹800$$

$$= \frac{43}{4} \times \frac{₹800}{1}$$

$$= \underline{₹8,600}$$

12.

$$1\frac{1}{4} + 2\frac{2}{3} = 3\frac{3+8}{12} = 3\frac{11}{12}$$

13.

Semisphere = $\frac{1}{2}$ of a circle

Area of semicircle = $\frac{1}{2}$ of Area of a circle

$$\text{i.e. } \frac{1}{2} \times \pi r^2 = \frac{\pi r^2}{2}$$

$$\text{diameter} = 2r$$

$$\therefore r = \frac{\text{diameter}}{2} = \frac{14}{2} = 7$$

$$\therefore \text{Area of semicircle} = \frac{\pi r^2}{2}$$

$$= \frac{\frac{22}{7} \times 7 \times 7}{2} = \frac{154}{2} = \underline{77 \text{ cm}^2}$$

14.

$$32.3\text{L}$$

$$7.66\text{L}$$

$$3.917\text{L}$$

$$22.000\text{L}$$

$$\underline{65.877\text{L}}$$

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